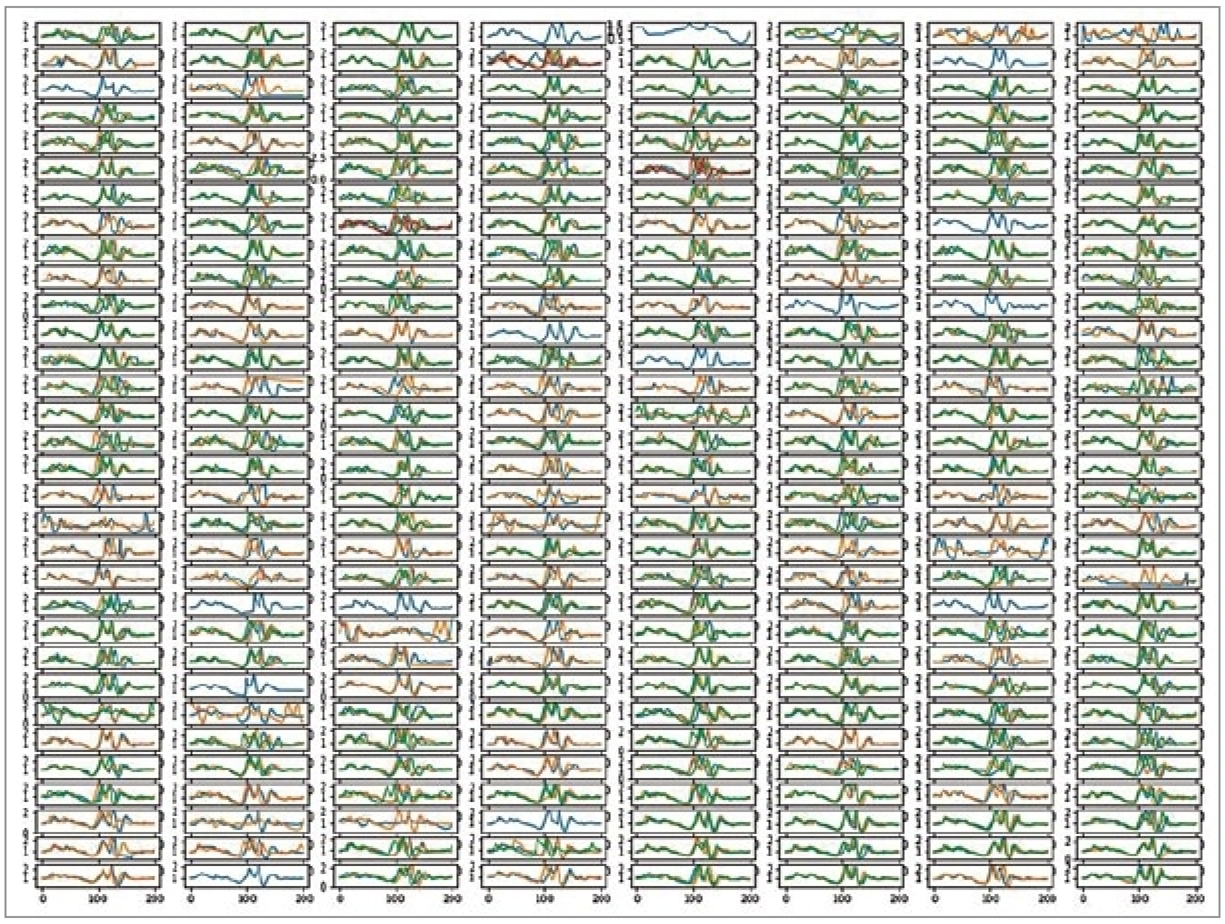

Visualisation of multi-session accelerometric gait cycle measurements from a volunteer

her pocket and walking in the comfort of his/her home.

In the figure, we see a visualisation of gait cycle time-series signals measured from the accelerometer of a smartphone carried by a volunteer at UnifyID labs.

As for the question of these measurements meeting the rigour of a proper medical examination, there exists a plethora of peer-reviewed published research that showcases that it is indeed the case.

Nearly eight years ago, Rigoberto et al presented a study that showcased how they had successfully used an off-the-shelf commercial smartphone as a tool to evaluate anticipatory postural adjustments before the beginning of normal gait in healthy subjects.

In the same year, Lemoyne et al demonstrated an implementation of an iPhone as a wireless accelerometer for rigourously quantifying gait characteristics.

Then, in 2012, Nishiguchi et al studied in detail the reliability and validity of gait analysis by Xperia SO-01B Android smartphone on 30 subjects. The scientists concluded that smartphone-based measurements indeed had the capacity to quantify gait parameters with a comparable degree of accuracy that was attained using a dedicated medical grade tri-axial accelerometeric sensor.

In 2016, we saw the first usage of a smartphone as a wireless gyroscopic platform for quantifying reduced arm swing in hemiplegie gait.

More recently, researchers from Toho university have developed Gait-Kun system. It consists of tri-axial accelerometers and shows promising levels of accuracy in measuring ataxia due to spino-cerebellar degeneration.

Geriatric care

A crucial component of delivering geriatric care is continuous monitoring. In this regard, the motion sensor-laden smartphone emerges as a key tool that can be used in lieu of, or in conjunction with, video monitoring systems.

One often-ignored facet of geriatric care is fall detection and response. As stated in a study on smartphone-based fall detection systems by Stefano Abbate et al, "A fall also has dramatic psychological consequences, since it drastically reduces the self-confidence and independence of affected people. This may contribute to future falls with more serious outcomes, or it may lead to a decline in health."

In 2012, He et al showcased how fall detection using smartphone motion sensors can be performed rather accurately, and how this technology can be a crucial piece in dramatically reducing the response time of healthcare professionals in the event of a fall.

In the same year, Fontecha et al demonstrated a system that could support physicians to construct a highly granular and centralised elderly frailty diagnosis, by munging activity data from an accelerometer-enabled cellphone with clinical data emanating from lab tests.

Besides this, Timed-Up-and-Go (TUG) test, which is also heavily used in geriatric care and for assessing the quality of recovery from hip fractures and cerebro-vascular accidents, can be performed with high accuracy using motion sensors on a smartphone.

In 2012, Mellone et al published a detailed study on the validity of a smartphone-based instrumented TUG test.

This was followed by another highly-cited work by Milosevic et al published in 2013, where the scientists introduced a smartphone application called sTUG that could completely automate the TUG test, so it could be performed by the patients themselves from the comfort of their homes.

Neurodegenerative disorders

Advancements in TUG test explored above are also applicable to neurodegenerative disorders. TUG test was initially designed for elderly persons, but is today also prescribed for patients with neurodegenerative disorders such as Parkinson's disease, multiple sclerosis, Alzheimer's disease and Huntington disease.

Besides this, there exists a fairly vast body of work where researchers have used motion sensors to help diagnose and assess neurodegenerative disorders. The first mobile gait analysis for early diagnosis and therapy monitoring in Parkinson's disease was published in 2011 by a group of German researchers at University Hospital of Erlangen.

Then, in 2012, Chung et al showcased how their stride detection algorithm, when applied to accelerometric data from commercial smartphones, could be used to perform early onset detection of Alzheimer's disease. This is based on the crucial observation that Alzheimer's patients exhibit a significantly shorter mean stride length and slower mean gait speed